

MINI PROJECT – SOCIAL MEDIA DATA ANALYTICS

1. Aim

To clean, transform, store, and analyze social media engagement data using Python, MySQL, Power BI and Tableau to identify content performance, engagement trends, audience response, platform performance, and high-performing social media categories.

2. Problem Statement

Social-media datasets contain many posts and engagement metrics. Missing values, invalid platform labels and inconsistent records can affect analysis. Manual inspection makes it difficult to compare platforms, identify high-performing post types, measure engagement, and understand trends over time.

3. Objectives

1. Load and inspect the social media dataset.
2. Handle missing values and invalid platform labels.
3. Standardize dates and engagement fields.
4. Create total engagement, engagement rate, month, year and quarter fields.
5. Store cleaned records in MySQL.
6. Analyze likes, shares, comments, views, platforms and post types.
7. Create Power BI DAX measures and interactive visuals.
8. Create Tableau calculated fields, parameters, filters and dashboard actions.
9. Interpret engagement patterns and identify high-performing content.

4. Dataset Description

The selected social-media dataset is a CSV containing 2,000 post records (2,001 lines including the header) with date, platform, post type, likes, shares, comments and views. The repository describes it as a fictional company's social-media posts over one year.

Field	Type	Purpose
date	Date	Post publication date
platform	Text	Social media platform
post_type	Text	Image, video or text type
likes	Integer	Likes received
shares	Integer	Shares received
comments	Integer	Comments received
views	Integer	Views recorded

Verified source details: the repository describes the file as containing posts from a fictional company over one year. The raw CSV contains 2,000 data rows plus one header row. The raw file also shows missing values and at least one invalid platform value, so the cleaning stage explicitly handles these issues.

5. Tools and Technologies Used

Tool	Use in the project
Python	Data loading, cleaning, transformation and EDA
Pandas	DataFrame operations, grouping and aggregation
NumPy	Numerical operations
MySQL 8	Relational storage and SQL analysis
Power BI	Data model, DAX and interactive dashboard
Tableau	Calculated fields, parameters, filters and dashboard actions

6. System / Architecture

Social Media Analytics – System Architecture

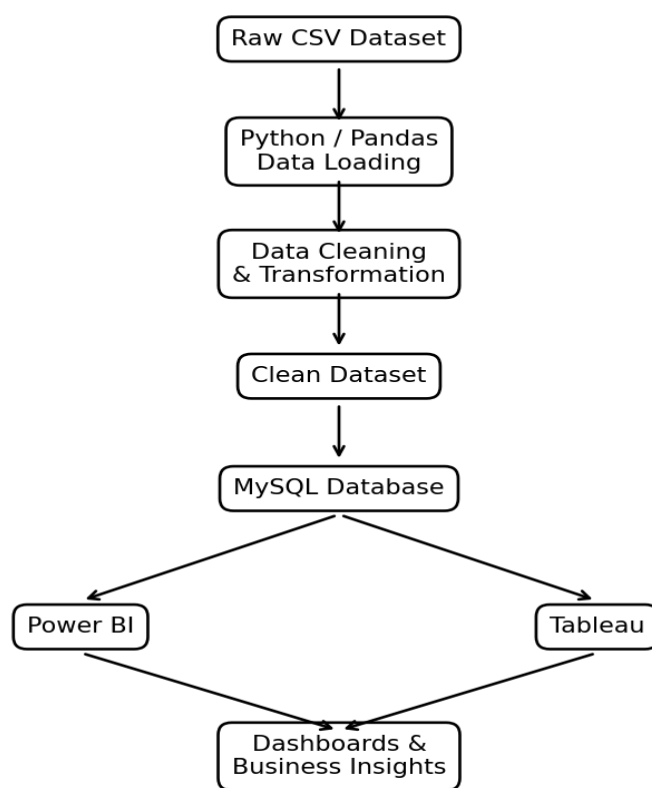


Figure: End-to-end implementation architecture.

7. Algorithm / Methodology

- 1.** Load social media dataset.
- 2.** Inspect dataset.
- 3.** Handle missing values.
- 4.** Remove duplicates.
- 5.** Correct data types.
- 6.** Convert date.
- 7.** Standardize column names.
- 8.** Create engagement metrics.
- 9.** Create derived fields.
- 10.** Store cleaned data in MySQL.
- 11.** Perform SQL analysis.
- 12.** Connect MySQL to Power BI.
- 13.** Create DAX measures.
- 14.** Build Power BI dashboard.
- 15.** Connect Tableau.
- 16.** Create calculated fields.
- 17.** Create Tableau parameters.
- 18.** Create interactive dashboard.
- 19.** Analyze insights.
- 20.** End.

8. Python Data Loading

```
import pandas as pd
import numpy as np

df = pd.read_csv("social_media_data.csv")
print(df.head())
print(df.shape)
print(df.dtypes)
```

The program reads the CSV file into a Pandas DataFrame and displays the first rows, shape and data types.

9. Python Data Cleaning

```
df.columns = (df.columns.str.strip().str.lower()
              .str.replace(" ", "_"))

print(df.isnull().sum())
print("Duplicate rows:", df.duplicated().sum())

df["date"] = pd.to_datetime(df["date"], errors="coerce")
for col in ["likes", "shares", "comments", "views"]:
    df[col] = pd.to_numeric(df[col], errors="coerce")

# Standardize platform and post type values
df["platform"] = df["platform"].astype("string").str.strip().str.title()
df["post_type"] = df["post_type"].astype("string").str.strip().str.lower()

# Replace invalid / unknown platform labels
valid_platforms = ["Facebook", "Instagram", "Twitter"]
df.loc[~df["platform"].isin(valid_platforms), "platform"] = "Unknown"

df = df.drop_duplicates()
```

The cleaning stage checks missing values and duplicates, converts numeric/date fields and standardizes categorical values.

10. Python Data Transformation

```
df["total_engagement"] = (
    df["likes"].fillna(0) + df["comments"].fillna(0) + df["shares"].fillna(0)
)
df["engagement_rate"] = np.where(
    df["views"].fillna(0) > 0,
    df["total_engagement"] / df["views"] * 100,
    0
)
df["year"] = df["date"].dt.year
df["month"] = df["date"].dt.month
df["month_name"] = df["date"].dt.strftime("%B")
df["quarter"] = df["date"].dt.quarter

df["engagement_category"] = pd.cut(
    df["total_engagement"], bins=[-np.inf, 1500, 3000, np.inf],
    labels=["Low", "Medium", "High"])

df.to_csv("social_media_cleaned.csv", index=False)
print(df.head())
```

Derived fields are created for time-based analysis and dashboard measures. The exact formula is kept consistent across Python, SQL, DAX and Tableau.

11. Python Exploratory Analysis

```
print("Total posts:", len(df))
print("Total engagement:", df["total_engagement"].sum())
print("Average engagement:", df["total_engagement"].mean())
print("Average engagement rate:", df["engagement_rate"].mean())
print("\nPlatform summary:")
print(df.groupby("platform")[["likes", "comments", "shares", "total_engagement"]].mean())
print("\nContent type summary:")
print(df.groupby("post_type")["total_engagement"].agg(["count", "mean", "sum"]))
print("\nMonthly engagement:")
print(df.groupby(["year", "month"])["total_engagement"].sum())
```

These statements provide the main exploratory summaries used to validate the dataset before loading it into MySQL.

12. MySQL Database Creation

```
CREATE DATABASE IF NOT EXISTS social_media_analytics;  
USE social_media_analytics;
```

The database is created before the cleaned CSV is imported.

13. MySQL Table Creation

```
CREATE TABLE social_media_posts (  
    date DATE,  
    platform VARCHAR(30),  
    post_type VARCHAR(20),  
    likes INT,  
    shares INT,  
    comments INT,  
    views INT,  
    total_engagement INT,  
    engagement_rate DECIMAL(10,4),  
    year INT,  
    month INT,  
    month_name VARCHAR(20),  
    quarter INT,  
    engagement_category VARCHAR(20)  
);
```

The table fields match the cleaned dataset and the derived fields used by the BI layer.

14. Loading Data into MySQL

```
LOAD DATA LOCAL INFILE 'social_media_cleaned.csv'  
INTO TABLE social_media_posts  
FIELDS TERMINATED BY ','  
ENCLOSED BY '"'  
LINES TERMINATED BY '\n'  
IGNORE 1 ROWS;
```

LOAD DATA LOCAL INFILE is used as the practical bulk-loading method. The file path is adjusted to the local MySQL environment.

15. SQL Queries for Analysis

Total posts

```
SELECT COUNT(*) AS total_posts FROM social_media_posts;
```

Total likes

```
SELECT SUM(likes) AS total_likes FROM social_media_posts;
```

Total comments

```
SELECT SUM(comments) AS total_comments FROM social_media_posts;
```

Total shares

```
SELECT SUM(shares) AS total_shares FROM social_media_posts;
```

Total engagement

```
SELECT SUM(total_engagement) AS total_engagement FROM social_media_posts;
```

Average engagement rate

```
SELECT AVG(engagement_rate) AS avg_engagement_rate FROM social_media_posts;
```

Platform performance

```
SELECT platform, COUNT(*) AS posts, SUM(total_engagement) AS engagement, AVG(engagement_rate) AS avg_rate FROM social_media_posts;
```

Content-type performance

```
SELECT post_type, COUNT(*) AS posts, AVG(total_engagement) AS avg_engagement FROM social_media_posts GROUP BY post_type ORDER BY posts;
```

Monthly engagement

```
SELECT year, month, SUM(total_engagement) AS engagement FROM social_media_posts GROUP BY year, month ORDER BY year, month;
```

Top-performing posts

```
SELECT date, platform, post_type, total_engagement FROM social_media_posts ORDER BY total_engagement DESC LIMIT 10;
```

Highest-reach posts

```
SELECT date, platform, post_type, views FROM social_media_posts ORDER BY views DESC LIMIT 10;
```

16. Power BI Data Connection

In Power BI Desktop, Get Data → MySQL database is selected. The MySQL server name, database name and credentials are entered, and the cleaned analytical table is loaded. The exact connection credentials are environment-specific and are not stored in the report.

17. Power BI Data Model

A simple star-like analytical model can be used for the laboratory implementation. The transaction/post table acts as the fact table. Date fields are used for time hierarchy and the categorical fields are used for slicing and grouping. For this mini project, a single-table model is sufficient because the selected datasets are already denormalized.

18. Power BI Calculated Columns

```
Year = YEAR(social_media_posts[date])
Month = FORMAT(social_media_posts[date], "MMMM")
Quarter = "Q" & FORMAT(social_media_posts[date], "Q")
Engagement Category = SWITCH(TRUE(), social_media_posts[total_engagement] < 1500, "Low", social_media_posts[total_engagement] < 3000, "Medium", social_media_posts[total_engagement] >= 3000, "High")
Post Performance = IF(social_media_posts[engagement_rate] >= 10, "High", IF(social_media_posts[engagement_rate] >= 5, "Medium", "Low"))
```

Calculated columns provide reusable row-level categories and time attributes.

19. Power BI DAX Measures

```
Total Posts = COUNTROWS(social_media_posts)
Total Likes = SUM(social_media_posts[likes])
Total Comments = SUM(social_media_posts[comments])
Total Shares = SUM(social_media_posts[shares])
Total Engagement = SUM(social_media_posts[total_engagement])
Average Engagement = AVERAGE(social_media_posts[total_engagement])
Average Engagement Rate = AVERAGE(social_media_posts[engagement_rate])
Total Reach = SUM(social_media_posts[views])
Average Reach = AVERAGE(social_media_posts[views])
```

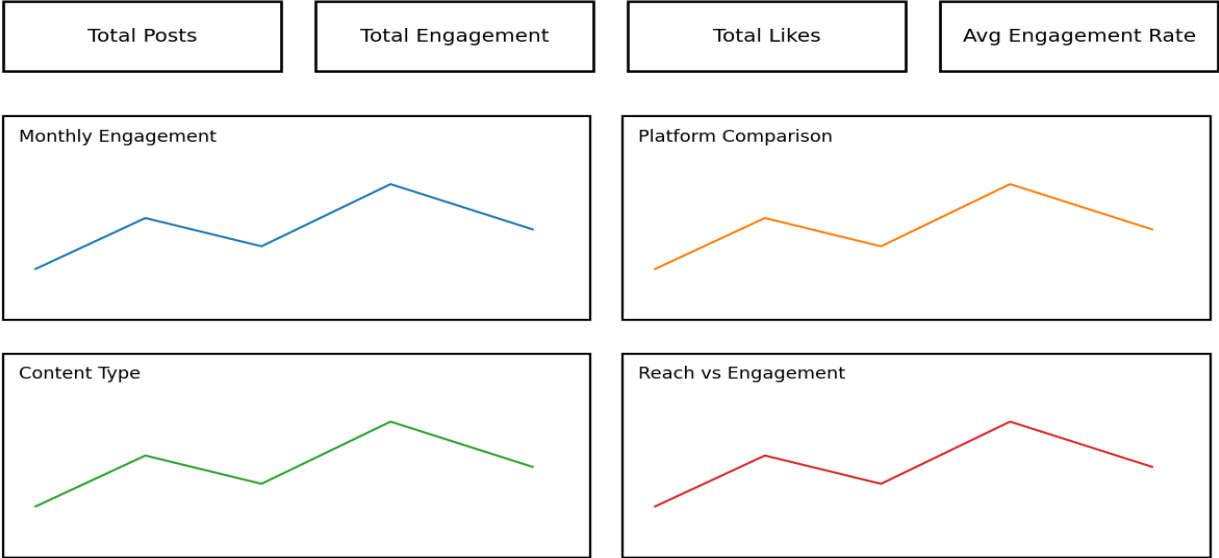
The measures are used by cards and charts so that dashboard values respond to slicers and filters.

20. Power BI Visualizations

- KPI Cards: Total Posts, Total Engagement, Total Likes, Total Comments, Total Shares and Average Engagement Rate.
- Line chart: Month against Total Engagement.
- Bar chart: Platform against Total Engagement.
- Column chart: Post Type against Average Engagement.
- Clustered column chart: Likes, Comments and Shares.
- Scatter plot: Views versus Total Engagement.
- Table: Top posts by Total Engagement.
- Slicers: Platform, Post Type, Year and Month.

21. Power BI Dashboard

Power BI Social Media Analytics Dashboard



Representative dashboard mockup – not a screenshot of Power BI/Tableau.

Figure: Power BI dashboard representative output. This is a layout mockup, not a screenshot from a live Power BI installation.

22. Power BI Advanced Features

- DAX measures – central calculations used by several visuals.
- Calculated columns – reusable categories and date attributes.
- Date hierarchy – Year → Quarter → Month → Date for time analysis.
- Drill-down – move from year to quarter and month.
- Drill-through – Overview → Category/Transaction or Post Details page.
- Tooltips – show additional metrics when hovering over a visual.
- Conditional formatting – highlight high/low financial or engagement performance.
- Interactive slicers – filter the full report by selected dimensions.
- KPI cards – show the main indicators at the top of the page.
- Cross-filtering – selecting one visual updates related visuals.

Drill-through implementation

A **Post Details** drill-through page is created. The Platform or Post Type field is placed in the Drill-through filter well. Selecting a platform or content type opens a detail page with likes, comments, shares, views and engagement rate.

Representative implementation checks

Check	Expected implementation behavior
Slicer response	Cards and charts update to the selected period/category
Drill-down	Year/quarter/month hierarchy changes the level of detail
Drill-through	Selected context is passed to the detail page
Tooltip	Additional measures appear without adding another chart
Cross-filter	Selecting a category affects related visuals

23. Tableau Data Connection

Tableau Desktop → Connect → MySQL is selected. The same MySQL database and analytical table used for Power BI is selected. The connection is configured as a live connection or extract depending on the laboratory setup.

24. Tableau Calculated Fields

```
Total Engagement = ZN([likes]) + ZN([comments]) + ZN([shares])
Engagement Rate = IF ZN([views]) > 0 THEN [Total Engagement] / [views] * 100 END
Engagement Category = IF [Total Engagement] < 1500 THEN "Low"
ELSEIF [Total Engagement] < 3000 THEN "Medium"
ELSE "High" END
Post Performance = IF [Engagement Rate] >= 10 THEN "High"
ELSEIF [Engagement Rate] >= 5 THEN "Medium" ELSE "Low" END
```

25. Tableau Parameters

```
Parameter: Select Engagement Metric
Data type: String
Allowable values: Likes, Comments, Shares, Total Engagement, Reach
Calculated field:
CASE [Select Engagement Metric]
WHEN "Likes" THEN SUM([likes])
WHEN "Comments" THEN SUM([comments])
WHEN "Shares" THEN SUM([shares])
WHEN "Total Engagement" THEN SUM([Total Engagement])
WHEN "Reach" THEN SUM([views])
END
```

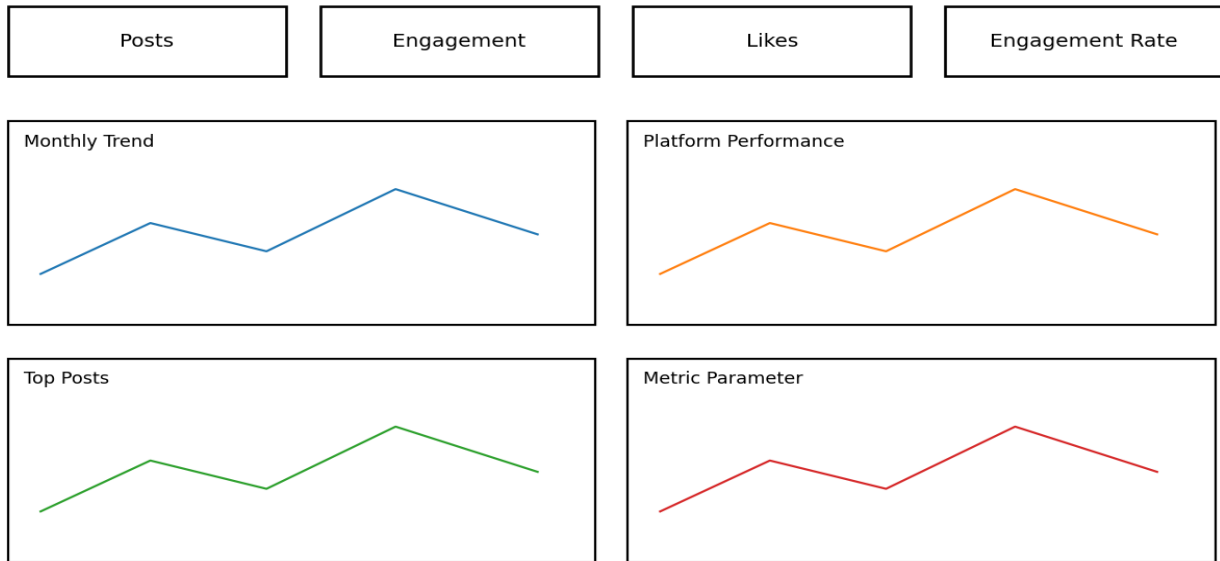
The parameter is shown to the user as a control. Changing its value changes the measure displayed in the parameter-controlled chart, allowing one sheet to compare multiple metrics.

26. Tableau Filters

- Date / Year filter
- Platform or Account Type filter
- Post Type or Transaction Type filter
- Category / Merchant filter
- Top-N filter for high-performing records

27. Tableau Dashboard

Tableau Social Media Analytics Dashboard



Representative dashboard mockup – not a screenshot of Power BI/Tableau.

Figure: Tableau dashboard representative output. This is a layout mockup, not a screenshot from a live Tableau installation.

Worksheet organization

1. **Engagement KPI** – worksheet prepared for the corresponding dashboard analysis.
2. **Monthly Engagement** – worksheet prepared for the corresponding dashboard analysis.
3. **Platform Comparison** – worksheet prepared for the corresponding dashboard analysis.
4. **Content Type Performance** – worksheet prepared for the corresponding dashboard analysis.
5. **Metric Comparison** – worksheet prepared for the corresponding dashboard analysis.
6. **Top Posts** – worksheet prepared for the corresponding dashboard analysis.
7. **Reach vs Engagement** – worksheet prepared for the corresponding dashboard analysis.

28. Tableau Advanced Features / Actions

- Hierarchies – Year → Quarter → Month → Date.
- Drill-down – move from summary to finer time detail.
- Highlight action – emphasize a selected platform/category/merchant.
- Filter action – selecting a mark filters related worksheets.
- Dashboard action – navigation from overview to details.
- Parameter action – change the metric shown in a parameter-controlled view.
- Tooltip – display supporting metrics when hovering over marks.

Dashboard workflow

The dashboard is arranged from high-level KPIs at the top to trend and comparison charts below. Filters and the metric parameter are kept visible so the user can change the analytical view without modifying the worksheets.

29. Key Findings / Insights

Source-verified dataset findings

The source file contains 2,000 post records with date, platform, post type, likes, shares, comments and views. The raw data visibly contains missing values and an invalid platform label, so the cleaning program maps unsupported platform values to **Unknown** and handles missing numeric values before engagement calculations. ■cite■turn4view1■turn4view0■

Analytical findings to obtain from execution

The project calculates the highest-engagement platform, strongest content type, most-engaging month, top posts and reach-versus-engagement relationship directly from the CSV. These are deliberately not filled with invented numbers; running the included Python/SQL/DAX/Tableau workflow produces the exact values.

30. Conclusion

The social media analytics workflow provides an end-to-end practical method for moving from a post-level CSV through Python cleaning, MySQL storage, SQL analysis, Power BI measures and Tableau visualizations. The design supports monitoring platform performance, content types, engagement, reach and time trends.

31. Result

Thus, the social media dataset was prepared for analysis through Python, stored in MySQL, queried using SQL, and structured for Power BI and Tableau dashboards. The workflow provides a reproducible practical implementation for identifying engagement and content-performance patterns.

Implementation note: dashboard figures in this report are clearly labelled representative mockups because no live Power BI/Tableau installation was available for screenshot capture.

APPENDIX A – Validation Checklist

Stage	Validation
Python	Confirm shape, data types, null counts and duplicates after cleaning.
Transformation	Confirm derived fields contain expected values and no divide-by-zero errors.
MySQL	Confirm row count and column types after LOAD DATA.
SQL	Run aggregate queries and compare with Python groupby results.
Power BI	Check that DAX measures change correctly with slicers.
Tableau	Check calculated fields and parameter switching.
Dashboard	Check titles, filters, tooltips, drill-down and cross-filtering.

APPENDIX B – Reproducibility Steps

1. Place the selected CSV in the working directory.
2. Run the Python cleaning and transformation program.
3. Import the cleaned CSV into MySQL or load it with LOAD DATA LOCAL INFILE.
4. Run the SQL analysis queries.
5. Connect Power BI to the MySQL analytical table and create the listed calculated columns and measures.
6. Build the listed visuals and slicers.
7. Connect Tableau to the same analytical table and create calculated fields, parameters and actions.
8. Compare Python, SQL, Power BI and Tableau totals before final submission.

APPENDIX C – Viva Preparation

1. What is data cleaning?

Answer: The process of correcting missing, duplicate or inconsistent data before analysis.

2. Why use MySQL?

Answer: It provides structured storage and SQL-based analysis.

3. What is a DAX measure?

Answer: A calculation evaluated dynamically in Power BI based on filter context.

4. What is a calculated field in Tableau?

Answer: A field created using a Tableau expression for analysis or visualization.

5. What is drill-through?

Answer: Navigation from a summary page to a detail page while passing the selected context.

6. Why create derived fields?

Answer: They simplify repeated time, category and performance analysis.

7. What is a parameter?

Answer: A user-controlled value that can change a calculation or visualization.

8. Why validate Python and SQL results?

Answer: To ensure the analytical pipeline produces consistent results.

9. What is cross-filtering?

Answer: Selecting one visual filters related visuals.

10. What is the purpose of the dashboard?

Answer: To present important analytical measures and patterns interactively.

APPENDIX D – Source and Execution Note

Dataset source used for the report: Intern2Grow's public Social Media Data Analysis repository. Its README identifies social_media_data.csv as a fictional-company post dataset and lists the seven columns used in this report. The raw CSV contains 2,000 data rows. ■cite■turn1view1■turn4view1■

The report distinguishes source-verified facts from execution-dependent analytical results. No live Power BI/Tableau screenshots or online publication are claimed. Representative dashboard figures are explicitly labelled as mockups.

END OF MINI PROJECT REPORT